

Nova Financial: Investment Analytics Dashboard

Problem Statement

You are building an **Investment Analytics Dashboard** for a portfolio management application. The dashboard fetches portfolio holding records, allows users to search and filter assets, and provides a currency display preference.

The application should use React hooks to manage state, side effects, and optimized financial calculations.

Task

Implement the solution inside the provided **App.jsx** file.

1. Data Fetching & Side Effects — **useEffect**

On component mount, simulate fetching portfolio data using **setTimeout**.

- After exactly **600ms**, populate the **assets** state with the provided **MOCK_ASSETS** array.
- Set **loading** to **false**.

Create a second **useEffect** that updates the browser title to:

Portfolio - Total: \${formattedTotal}

The **formattedTotal** should be the total value calculated from the current filtered assets.

2. Optimized Computations — **useMemo**

Implement a single `useMemo` that returns:

```
{
  filteredAssets,
  totalValue
}
```

Filter the assets based on:

- `searchTerm` — asset name matching should be **case-insensitive**.
- `selectedCategory` — filter by `assetClass`.
- If `selectedCategory` is "All", do not apply the category filter.

Calculate `totalValue` by summing:

`quantity × currentPrice`

for all filtered assets.

Inside the `useMemo` callback, include:

```
window.analyticsMemoCount = (window.analyticsMemoCount || 0) + 1;
```

The `useMemo` dependency array should contain only:

```
[assets, searchTerm, selectedCategory]
```

Changing `currency` should not trigger the `useMemo`.

3. Component Composition

FilterBar

Pass the following props to `<FilterBar />`:

```
searchTerm
setSearchTerm
selectedCategory
setSelectedCategory
currency
```

setCurrency

PortfolioMetrics

Pass the following props to `<PortfolioMetrics />`:

totalValue
currency
assetCount

Where:

assetCount = filteredAssets.length

AssetTable

Pass the following props to `<AssetTable />`:

assets = filteredAssets
currency = currency

4. Provided Mock Data

Use the following `MOCK_ASSETS` array:

```
const MOCK_ASSETS = [  
  {  
    id: 1,  
    name: "Apple Inc.",  
    symbol: "AAPL",  
    assetClass: "Stocks",  
    quantity: 10,  
    currentPrice: 195.5,  
  },  
  {  
    id: 2,  
    name: "Microsoft Corp.",  
    symbol: "MSFT",  
    assetClass: "Stocks",
```

```
    quantity: 8,  
    currentPrice: 420.25,  
  },  
  {  
    id: 3,  
    name: "Bitcoin",  
    symbol: "BTC",  
    assetClass: "Crypto",  
    quantity: 0.5,  
    currentPrice: 65000,  
  },  
  {  
    id: 4,  
    name: "Ethereum",  
    symbol: "ETH",  
    assetClass: "Crypto",  
    quantity: 2,  
    currentPrice: 3500,  
  },  
  {  
    id: 5,  
    name: "US Treasury Bond",  
    symbol: "UST",  
    assetClass: "Bonds",  
    quantity: 20,  
    currentPrice: 102.5,  
  },  
  {  
    id: 6,  
    name: "Gold ETF",  
    symbol: "GLD",  
    assetClass: "Commodities",  
    quantity: 15,  
    currentPrice: 220.75,  
  },  
];
```

5. Initial State

Use the following initial values:

```
assets: []  
loading: true  
searchTerm: ""  
selectedCategory: "All"  
currency: "USD"
```

Asset Categories

All
Stocks
Crypto
Bonds
Commodities

Currency Options

USD
EUR
INR

Components

The dashboard should contain:

```
App  
├── FilterBar  
├── PortfolioMetrics  
└── AssetTable
```

All implementation should be done in **App.jsx** using the provided project structure and components.